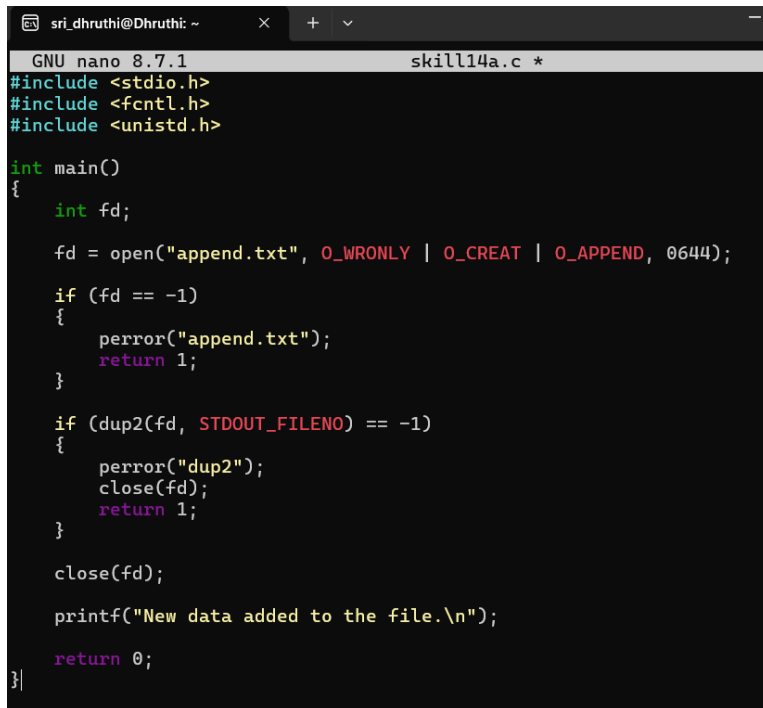


OS Skill 14

14(a) Append Redirection

What it does: Opens a file in append mode and redirects output to it without deleting the existing data.



```
GNU nano 8.7.1 skill14a.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;

    fd = open("append.txt", O_WRONLY | O_CREAT | O_APPEND, 0644);

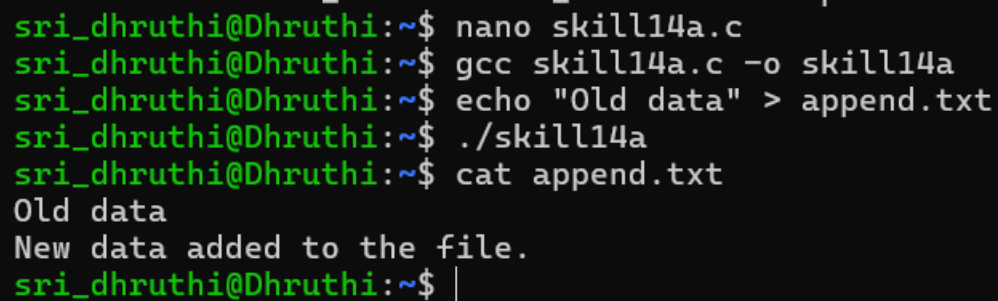
    if (fd == -1)
    {
        perror("append.txt");
        return 1;
    }

    if (dup2(fd, STDOUT_FILENO) == -1)
    {
        perror("dup2");
        close(fd);
        return 1;
    }

    close(fd);

    printf("New data added to the file.\n");

    return 0;
}
```



```
sri_dhruthi@Dhruthi:~$ nano skill14a.c
sri_dhruthi@Dhruthi:~$ gcc skill14a.c -o skill14a
sri_dhruthi@Dhruthi:~$ echo "Old data" > append.txt
sri_dhruthi@Dhruthi:~$ ./skill14a
sri_dhruthi@Dhruthi:~$ cat append.txt
Old data
New data added to the file.
sri_dhruthi@Dhruthi:~$ |
```

14(b) stderr Redirection

What it does: Redirects error messages from stderr into a file so they can be captured and checked.

```
sri_dhruthi@Dhruthi: ~
GNU nano 8.7.1 skill14b.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;

    fd = open("error.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1)
    {
        perror("error.txt");
        return 1;
    }

    if (dup2(fd, STDERR_FILENO) == -1)
    {
        close(fd);
        return 1;
    }

    close(fd);

    fprintf(stderr, "Error: File could not be opened.\n");
    fprintf(stderr, "Error: Invalid input detected.\n");

    return 0;
}
```

```
sri_dhruthi@Dhruthi:~$ nano skill14b.c
sri_dhruthi@Dhruthi:~$ gcc skill14b.c -o skill14b
sri_dhruthi@Dhruthi:~$ ./skill14b
sri_dhruthi@Dhruthi:~$ cat error.txt
Error: File could not be opened.
Error: Invalid input detected.
sri_dhruthi@Dhruthi:~$ |
```